

Week 5 — Session 2

Course: 10211CS225 — Problem Solving and Testing using Java

Unit II: Algorithmic Thinking & Competitive Problem Patterns

Topic: Kadane's Algorithm

Language: Java

Focus: Master Kadane's algorithm and extend it using the total-sum/minimum-subarray idea for circular arrays.

Task 1: Maximum Subarray

Practice: <https://www.hackerrank.com/challenges/maxsubarray/>

Complete Java Solution

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int t=sc.nextInt();

        while(t-->0){
            int n=sc.nextInt();
            long[] a=new long[n];
            for(int i=0;i<n;i++) a[i]=sc.nextLong();

            long cur=a[0], best=a[0];
            long positive=a[0]>0?a[0]:0;
            long maxElement=a[0];

            for(int i=1;i<n;i++){
                cur=Math.max(a[i],cur+a[i]);
                best=Math.max(best,cur);
                maxElement=Math.max(maxElement,a[i]);
                if(a[i]>0) positive+=a[i];
            }

            long subsequence=positive>0?positive:maxElement;
            System.out.println(best+" "+subsequence);
        }
        sc.close();
    }
}
```

Task 2: Maximum Sum Circular Subarray

Practice: <https://leetcode.com/problems/maximum-sum-circular-subarray/>

Complete Java Solution

```
import java.util.*;

public class Main {
    static int maxCircular(int[] a){
        int total=a[0], maxEnd=a[0], max=a[0];
        int minEnd=a[0], min=a[0];

        for(int i=1;i<a.length;i++){
            total+=a[i];
            maxEnd=Math.max(a[i],maxEnd+a[i]);
            max=Math.max(max,maxEnd);
            minEnd=Math.min(a[i],minEnd+a[i]);
            min=Math.min(min,minEnd);
        }
        if(max<0) return max;
        return Math.max(max,total-min);
    }

    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] a=new int[n];
        for(int i=0;i<n;i++) a[i]=sc.nextInt();
        System.out.println(maxCircular(a));
        sc.close();
    }
}
```

Programs are standalone Java solutions. Java 17+ is recommended.